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The equation $6 \times 5 - 4 \div 2 - 3 + 7)^0 - 1)^2 = 6$ is missing two opening parentheses. It can be made true if opening parentheses are placed immediately before each of two of the numbers.

Assume that standard order-of-operations principles are followed, whereby multiplication and division are performed before addition and subtraction, unless parentheses indicate otherwise. Select in the table a number before which the first opening parenthesis could be placed and a number before which the second opening parenthesis could be placed, such that the equation would express a true statement. Make only two selections, one in each column.

First parenthesis immediately before	Second parenthesis immediately before	
		2
		3
		4
		5
		6

Answer Explanation:

Apply

Try all possible placements of the two opening parentheses:

Before 5 and 4 $6 \times (5 - (4 \div 2 - 3 + 7)^0 - 1)^2 = 54$

Before 5 and 2 $6 \times (5 - 4 \div (2 - 3 + 7)^0 - 1)^2 = 0$

Before 5 and 3 $6 \times (5 - 4 \div 2 - (3 + 7)^0 - 1)^2 = 6$

Before 5 and 7 $6 \times (5 - 4 \div 2 - 3 + (7)^0 - 1)^2 = 0$

Before 5 and 1 $6 \times (5 - 4 \div 2 - 3 + 7)^0 - (1)^2 = 5$

Before 4 and 2 $6 \times 5 - (4 \div (2 - 3 + 7)^0 - 1)^2 = 21$

Before 4 and 3 $6 \times 5 - (4 \div 2 - (3 + 7)^0 - 1)^2 = 30$

Before 4 and 7 $6 \times 5 - (4 \div 2 - 3 + (7)^0 - 1)^2 = 29$

Before 4 and 1 $6 \times 5 - (4 \div 2 - 3 + 7)^0 - (1)^2 = 28$

Before 2 and 3 $6 \times 5 - 4 \div (2 - (3 + 7)^0 - 1)^2 =$ undefined division by 0

Before 2 and 7 $6 \times 5 - 4 \div (2 - 3 + (7)^0 - 1)^2 = 26$

Before 2 and 1 $6 \times 5 - 4 \div (2 - 3 + 7)^0 - (1)^2 = 25$

Before 3 and 7 $6 \times 5 - 4 \div 2 - (3 + (7)^0 - 1)^2 = 10$

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Before 3 and 7 $6 \times 5 - 4 \div 2 - (3 + (7)^0 - 1)^2 = 10$

Before 3 and 1 $6 \times 5 - 4 \div 2 - (3 + 7)^0 - (1)^2 = 26$

Before 7 and 1 $6 \times 5 - 4 \div 2 - 3 + (7)^0 - (1)^2 = 25$

***First parenthesis immediately before*:**

The third row of the table shows an expression with value 6. The first opening parenthesis is before 5.

The correct answer is *5*.

***Second parenthesis immediately before*:**

The third row of the table shows an expression with value 6. The second opening parenthesis is before 3.

The correct answer is *3*.

•

Historian: New York City became an influential art center in the twentieth century, as seen, for example, in the influential 1913 Armory Show and the growth of museums. This was due primarily to a combination of New York's large and varied population, its basic resources, and its access to European art trends. The city could never have become an influential art center without the era's increased local wealth, which allowed more people to buy art. However, the changes were driven principally by the diversity of the city's population, including intellectuals and artists from Europe, and a culture that prized energy and creativity, which stimulated the artistic endeavors of many people.

Select for *Precondition* the statement that the passage most strongly suggests describes a precondition for New York becoming an influential art center. And select for *Strengthener* the statement that would, if true, most clearly strengthen the support for the historian's explanation of why the city became an influential art center. Make only two selections, one in each column.

Precondition	Strengthener	
		New York showed significant economic development leading up to the early twentieth century.
		New York afforded easy access to major art museums in other cities in the United States.
		Members of the Hudson River School of painters helped promote art in New York.
		The Armory Show and other influential New York shows like it featured many prominent European artists.
		The composition of European groups immigrating to New York changed in the early 1900s.

Answer Explanation:

Evaluate

***Precondition*:**

The first answer choice is strongly suggested as a precondition in the passage's third sentence: "The city could never have become an influential art center without the era's increased local wealth, which allowed more people to buy art." Also, none of the other answer choices strongly suggests such a precondition—the second and third answer choices involve issues not mentioned in the passage, the fourth answer choice is a misrepresentation of an aftereffect, and, at best, the fifth answer choice is only suggested as one of the reasons for the changes that led New York to become an influential art center.

The correct answer is *New York showed significant economic development leading up to the early 20th century*.

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The correct answer is *New York showed significant economic development leading up to the early 20th century.*

***Strengthener*:**

The first and fifth answer choices are arguably included in the passage, and thus would essentially add nothing to strengthen the historian's explanation. Although the second and third answer choices might provide some strengthening of the historian's explanation, neither is particularly relevant to the issues that are actually mentioned in the passage. The fourth answer choice mentions "prominent European artists" in a way that supports the passage's linking of them to New York's economic development.

The correct answer is *The Armory Show and other influential New York shows like it featured many prominent European artists.*

•

A farmer wants to allocate her expenditures to maximize her profits during the coming year. Let C be her spending on cropland, F her spending on fertilizer, L her spending on labor, M her spending on machinery, and S her spending on seed. She has estimated that, if she allocates at least 100 euros to each of these categories, then her profit during the coming year will be $(C-97)(L-92)(S-95)(M-87)(F-90)$ euros. She wishes to calculate the benefits of expenditure greater than 100 euros in one of these categories, if she were to spend exactly 100 euros in each of the others.

Select for *Greatest returns* the resource for which, according to the desired calculation from the model, each euro of spending beyond 100 euros is predicted to make the greatest contribution to profits (or least contribution to losses). And select for *Least returns* the resource for which, according to the desired calculation, each euro of spending beyond the minimum necessary amount is predicted to make the least contribution to profits (or greatest contribution to losses). Make only two selections, one in each column.

Greatest returns	**Least returns**	
		Cropland
		Fertilizer
		Labor
		Machinery
		Seed

Answer Explanation:

Strategize

Let $P=(C-97)(L-92)(S-95)(M-87)(F-90)$. Let x represent the amount by which an expenditure is greater than 100 euros. The following table summarizes the calculations for P given certain constraints. For example, if $C=100+x$ and $L=S=M=F=100$, then
$P=(100+x-97)(100-92)(100-95)(100-87)(100-90)=$
$(3+x)(8)(5)(13)(10)=(3+x)(5{,}200)=15{,}600+5{,}200x$.

C	$100+x$	100	100	100	100
L	100	$100+x$	100	100	100
S	100	100	$100+x$	100	100
M	100	100	100	$100+x$	100
F	100	100	100	100	$100+x$

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F	100	100	100	100	$100 + x$
P	$15{,}600 + 5{,}200x$	$15{,}600 + 1{,}950x$	$15{,}600 + 3{,}120x$	$15{,}600 + 1{,}200x$	$15{,}600 + 1{,}560x$

From the table, for every euro spent beyond 100 euros, cropland increases profit by 5,200 euros, labor by 1,950 euros, seed by 3,120 euros, machinery by 1,200 euros, and fertilizer by 1,560 euros.

***Greatest returns*:**

The greatest of these increases in profits for every euro spent beyond 100 euros is for cropland.

The correct answer is *Cropland*.

***Least returns*:**

The least of these increases in profits for every euro spent beyond 100 euros is for machinery.

The correct answer is *Machinery*.

•

At a certain factory, 4 processes—A, B, C, and D—are carried out 24 hours per day, 7 days per week. Process A operates on a 60-hour cycle; that is, Process A takes 60 hours to complete and immediately begins again when it is completed. Likewise, Process B operates on a 24-hour cycle, Process C operates on a 27-hour cycle, and Process D operates on a 9-hour cycle. On Monday, all 4 processes began together at 10:00 in the morning.

On the basis of the information provided, select for *Processes A, B, and D* the day of the week on which Processes A, B, and D will next begin together at 10:00 in the morning. Also select for *All 4 processes* the day of the week on which Processes A−D will next begin together at 10:00 in the morning. Make only two selections, one in each column.

Processes A, B, and D	**All 4 processes**	
		Monday
		Tuesday
		Wednesday
		Thursday
		Friday

Answer Explanation:

Apply

***Processes A, B, and D*:**

Processes A, B, and D will all end at 10:00 a.m. on the same day and therefore will begin together at 10:00 a.m. the next time when the time elapsed since they started together is the least common multiple of 60, 24, and 9. This can be found by analyzing the prime factorizations of 60, 24, and 9, which are $2^2 \times 3 \times 5$, $2^3 \times 3$, and 3^2, respectively. The least common multiple is the product of the greatest power of each of the prime factors in the three numbers, so the least common multiple of 60, 24, and 9 is $2^3 \times 3^2 \times 5 = 360$. Thus, the three processes will begin at the same time 360 hours, which is $\frac{360}{24} = 15$ days, after they began together; 15 days (or 2 weeks and 1 day) after Monday is Tuesday.

The correct answer is *Tuesday*.

***All 4 processes*:**

Processes A, B, C, and D will all end at 10:00 a.m. on the same day and therefore will begin together at 10:00 a.m. the next time when the time elapsed since they started together is the least common multiple of 60, 24, 27, and 9. It has already been determined that the least common multiple of 60, 24, and 9 is $2^3 \times 3^2 \times 5 = 360$. The prime

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has already been determined that the least common multiple of 60, 24, and 9 is $2^3 \times 3^2 \times 5 = 360$. The prime factorization of 27 is 3^3, so any multiple of 27 must have 3 factors of 3. Thus, the least common multiple of 60, 24, 27, and 9 is $2^3 \times 3^3 \times 5 = 1{,}080$, which is $\frac{1{,}080}{24} = 45$ days; 45 days (or 6 weeks and 3 days) after Monday is Thursday.

The correct answer is *Thursday*.

•

An amateur athletic team has the following requirement: "Each athlete who fails to return to his or her designated room prior to a designated time will be suspended for the next athletic event." The requirement was violated by the team's best athlete on the night before a prominent athletic event.

Assistant Coach A: The only reason to not suspend the athlete is if the athlete broke the requirement because of factors outside the athlete's control. Since that was not the reason, no exception should be made for this athlete.

Assistant Coach B: I know that we would normally punish the athlete with a suspension, but doing so in this case would harm the team. I think we should punish the athlete in another way that would not harm the team.

Select for *Supports Assistant Coach A* the principle that most strongly supports the reasoning expressed by Assistant Coach A, and select for *Supports Assistant Coach B* the principle that most strongly supports the reasoning expressed by Assistant Coach B. Make only two selections, one in each column.

Supports Assistant Coach A	**Supports Assistant Coach B**	
		Amateur athletic teams should not punish elite team members who break team requirements.
		The punishment of a team member for breaking a team requirement should be based on the likely outcomes for the team rather than the punishment specified for breaking the requirement.
		The punishment specified for breaking a team requirement should always be administered in order to ensure that all team members are treated equally.
		If an athlete is reasonably able to fulfill a team requirement, then the punishment specified for breaking the requirement should be administered.
		If a punishment of a team member specified by a team requirement would harm the team, then the team member should not be punished.

Answer Explanation:

Evaluate

***Supports Assistant Coach A*:**

Assistant Coach A claims that it was not because of anything outside the control of the athlete in question that the athlete broke the requirement, and, as a result, there should be no exception made for the athlete. What would help support this thinking? Suppose that the following principle is true: If an athlete is reasonably able to fulfill any particular team requirement, then the punishment specified for breaking the requirement should be administered. Assistant Coach A claims that it was not because of anything outside the athlete's control that the athlete broke the requirement. It follows, then, that the athlete should have been able to fulfill that team requirement, and yet the athlete didn't. This principle, then, would support Assistant Coach A's conclusion that there should be no exception made for the athlete.

The correct answer is *If an athlete is reasonably able to fulfill a team requirement, then the punishment specified for breaking the requirement should be administered.*

***Supports Assistant Coach B*:**

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***Supports Assistant Coach B*:**

Assistant Coach B states that punishing the athlete in question with a suspension would harm the team, so the athlete should be punished in another way that would not harm the team. Suppose that the following principle is true: The punishment of a team member for breaking a team requirement should be based on the likely outcomes for the team rather than the punishment specified for breaking the requirement. Assistant Coach B claims that the punishment specified by the requirement would harm the team. Therefore, assuming the stated principle is true, and the reasonable assumption that the team should not be harmed just because one member of the team violated a requirement, Assistant Coach B is correct that the athlete should be punished in some other way that would not harm the team.

The correct answer is *The punishment of a team member for breaking a team requirement should be based on the likely outcomes for the team rather than the punishment specified for breaking the requirement.*

•

Rashika is beginning a new job in human resources and expects to be assigned to several of the many human resources work groups. She has been told that there is a probability of 12 that she will be assigned to the Recruitment Work Group and that there is a probability of 34 that she will be assigned to the New Technology Work Group. She has no other information about the probabilities of the various possible assignments, or whether her assignment to one of these groups affects her chances of assignment to the other.

Select for *Greatest probability for both assignments* the greatest probability, compatible with the given probabilities, that Rashika will be assigned to both the Recruitment Work Group and the New Technology Work Group. And select for *Least probability for both assignments* the least probability, compatible with the given probabilities, that Rashika will be assigned to both of these groups. Make only two selections, one in each column.

Greatest probability for both assignments	**Least probability for both assignments**	
		0
		$\frac{1}{4}$
		$\frac{1}{2}$
		$\frac{3}{4}$
		1

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Answer Explanation:

Infer

***Greatest probability for both assignments*:**

In terms of the probability x that Rashika will be assigned to both groups, consider the probabilities involved. For example, the probability that Rashika will be assigned to the Recruitment Work Group and not be assigned to the New Technology Work Group is $\left(\frac{1}{2} - x\right)$, and the probability that Rashika will be assigned to the Recruitment Work Group is $\left(\frac{1}{2} - x\right) + x = \frac{1}{2}$. The two constraints that follow give all restrictions on the values that can appear in the regions in this graph: (1) Each value is between 0 and 1, inclusive. (2) The sum of the three values is between 0 and 1, inclusive. In particular, from (1) we have $\frac{1}{2} - x \geq 0$, or $x \leq \frac{1}{2}$. Moreover, since $x = \frac{1}{2}$ satisfies both (1) and (2), it follows that the greatest possible value of x is $\frac{1}{2}$.

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The correct answer is $\frac{1}{2}$.

Least probability for both assignments:

In the explanation above, from (2) we have $\left(\frac{1}{2}-x\right)+x+\left(\frac{3}{4}-x\right)\leq 1$, or $x\geq\frac{1}{4}$. Moreover, since $x=\frac{1}{4}$ satisfies both (1) and (2), it follows that the least possible value of x is 14.

The correct answer is $\frac{1}{4}$.

•

In a study conducted over several years, seabird and domesticated cat populations on a geographically isolated island changed from year to year. Researchers found that over the course of the study, the relationship (R) between seabirds and domesticated cats was such that the island's seabird population was three times as likely to decrease from the previous year if the island's domesticated cat population increased (even if slightly) during the same year. The researchers are about to begin a second, follow-up study with the same duration as the first study. Based on recent trends, the researchers made the following projections: R will hold and the island's domesticated cat population will decrease during more years of the second study than it did in their first study.

Assuming that the information above is true, select for *Can be inferred as true* the statement that can be most reasonably inferred as true from the information provided, and select for *Can be inferred as false* the statement that can be most reasonably inferred as false from the information provided. Make only two selections, one in each column.

Can be inferred as true	**Can be inferred as false**	
		If the researchers' projections are accurate, the island's seabird population is likely to increase during more years of the second study than it did in the first.
		During the first study, most years had an increase in the island's domesticated cat population.
		If the researchers' projections are accurate, the island's seabird population is likely to increase during most of the years of the second study.
		During the first study, the island's seabird population decreased only when the island's domesticated cat population increased.
		During the first study, most years during which the island's seabird population decreased were years during which the island's domesticated cat population increased.

Answer Explanation:

Evaluate

***Can be inferred as true*:**

In each year of a study that was conducted over a period of several years, the populations of both seabirds and domesticated cats on a geographically isolated island changed—that is, the populations either increased or decreased each year from the previous year's numbers. The researchers found that a certain relationship R held: The island's seabird population was three times as likely to decrease from the previous year if the island's domesticated cat population increased even slightly during the same year. Because the populations of each of the two types of animals changed each year, we can infer that when the domesticated cat population decreased, the seabird population was only one-third as likely to increase as when the cat population decreased. It also follows, then, that the seabird

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population was more likely to increase when the cat population decreased than when it increased. So, if the researchers are correct that R will hold during the second, follow-up study that will last the same number of years as the first study, and that the domesticated cat population will decrease during more years than it did during the first study, then it can reasonably be inferred that the seabird population will increase in more years than it did in the first study.

The correct answer is *If the researcher's projections are accurate, the island's seabird population is likely to increase during more years of the second study than it did in the first.*

***Can be inferred as false*:**

If it was determined in the first study that the island's seabird population was three times as likely to decrease in years when the island's domesticated cat population increased, then it cannot be true that the island's seabird's population *never* decreased in years when the island's domesticated cat population did not increase. To see this, consider that for R to hold, the number of years x in which the island's cat population increased and the island's seabird population decreased has to be three times as great as the number of years y in which the island's cat population did *not* increase and the island's seabird population decreased. In other words, $\frac{x}{y} = 3$. And if $\frac{x}{y} = 3$, then y cannot equal 0.

Therefore, it must NOT be the case that, during the first study, the island's seabird population decreased only when the island's cat population increased.

The correct answer is *During the first study, the island's seabird population decreased only when the island's cat population increased.*

•

According to a prominent investment adviser, Company X has a 50% chance of posting a profit in the coming year, whereas Company Y has a 60% chance of posting a profit in the coming year.

Select for *Least probability for both* the least probability, compatible with the probabilities provided by the investment adviser, that both Company X and Company Y will post a profit in the coming year. And select for *Greatest probability for both* the greatest probability, compatible with the probabilities provided by the investment adviser, that both Company X and Company Y will post a profit in the coming year. Make only two selections, one in each column.

Least probability for both	Greatest probability for both	
		5%
		10%
		25%
		50%
		60%
		80%

Answer Explanation:

Infer

***Least probability for both*:**

To determine what the least probability is of both Company X and Company Y posting a profit in the coming year, determine what is the greatest probability of one of the companies—say Company X—posting a profit and the other company—Company Y—not posting a profit, and then subtract that result from the probability of Company X posting a profit. Because the probability of Company Y posting a profit in the coming year is 60%, then the probability of Company Y ***not*** posting a profit in the coming year is 40%. The greatest probability of Company X posting a profit and Company Y ***not*** posting a profit in the coming year cannot exceed the lesser of the probability of

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posting a profit and Company Y not posting a profit in the coming year cannot exceed the lesser of the probability of Company X posting a profit in the coming year—which is 50%—and the probability of Company Y ***not*** posting a profit in the coming year—which is 40%. Therefore, at most, the probability of both Company X posting a profit but Company Y ***not*** posting a profit in the coming year is 40%. Subtracting the 40% probability of Company X posting a profit from the probability of Company Y ***not*** posting a profit from the 50% probability of Company X posting a profit gives us the least probability of both Company X and Company Y posting a profit in the coming year, namely 10%.

The correct answer is *10%*.

***Greatest probability for both*:**

The greatest probability of both Company X and Company Y posting a profit in the coming year cannot exceed the lesser of the probability of Company X posting a profit in the coming year—which is 50%—and the probability of Company Y posting a profit in the coming year—which is 60%. Therefore, the greatest probability of both Company X and Company Y posting a profit in the coming year is 50%.

The correct answer is *50%*.

•

Lee is planning a trip and estimates that, rounded to the nearest 5 kilometers (km), the length of the trip will be 560 km and that, rounded to the nearest 14 hours, the driving time for the trip will be 7 hours. If these estimates are correct, then Lee's average driving speed during the trip will be between ___x___ kilometers per hour and ___y___ kilometers per hour, where $x < y$.

From the values given in the table, select for x and for y the values that complete the statement in such a way that the interval between the selected values includes all possible average speeds for Lee's trip and $y - x$ is minimal. Make only two selections, one in each column.

x	y	
		73
		76
		79
		82
		85

Answer Explanation:

Infer

The least distance that rounds to 560 kilometers (km) when rounding to the nearest 5 km is 557.5 km and the greatest is 562.5 km. The least length of time that rounds to 7 hours when rounding to the nearest $\frac{1}{4}$ hour is $6\frac{7}{8} = 6.875$ hours and the greatest is $7\frac{1}{8} = 7.125$ hours.

x:

The least possible average speed is $\frac{557.5}{7.125} \approx 78.25$. Since the value of x must be less than 78.25, it follows that 73 and 76 are the only possible values. Because $y - x$ must be minimal, $x = 76$.

The correct answer is *76*.

y:

The greatest possible average speed is $\frac{562.5}{6.875} \approx 81.82$. Since the value of y must be greater than 81.82, it follows

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that 82 and 85 are the only possible values. Because $y - x$ must be minimal, $y = 82$.

The correct answer is *82*.

•

A certain theater has 500 seats. Some are on the main floor and sell for $50 each; some are in the first balcony and sell for $45 each; and the rest are in the second balcony and sell for $35 each. When all of the seats are sold for a performance, the gross revenue for that performance is $20,900. Of the three seating areas, the second balcony has the most seats.

Select for *First balcony* a number of seats in this theater's first balcony, and select for *Second balcony* a number of seats in this theater's second balcony such that the selections are jointly consistent with the information provided. Make only two selections, one in each column.

First balcony	Second balcony	
		100
		150
		190
		210
		250
		300

Answer Explanation:

Infer

If x represents the number of seats in the first balcony and y represents the number of seats in the second balcony, then $500 - (x + y) = 500 - x - y$ represents the number of seats on the main floor. The following equation represents the given information.

$$\begin{aligned} 50(500 - x - y) + 45x + 35y &= 20{,}900 \\ 25{,}000 - 50x - 50y + 45x + 35y &= 20{,}900 \\ -5x - 15y &= -4{,}100 \\ x + 3y &= 820 \end{aligned}$$

Since the second balcony has the most seats, it follows that $y = 210$ or $y = 250$ or $y = 300$. If $y = 210$, then $x = 190$. If $y = 250$, then $x = 70$, which is not among the answer choices. If y = 300, then $x = -80$, which is not a valid value for x. Therefore, the first balcony has 190 seats, and the second balcony has 210 seats.

***First balcony*:**

The correct answer is *190*.

***Second balcony*:**

The correct answer is *210*.

•

A company purchased at least 10 units of a certain product from each of Suppliers X and Y. Supplier X charged a $100 fixed cost, $20 per unit for each of the first 10 units, and $5 per unit for each unit purchased in excess of 10 units. Supplier Y charged a $150 fixed cost, $15 per unit for each of the first 10 units, and $10 per unit for each unit purchased in excess of 10 units. Including the fixed costs, the combined average cost for the units was $20 per unit.

Select a *number of units purchased from Supplier* X and a *number of units purchased from Supplier Y* that are jointly

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Select a *number of units purchased from Supplier X* and a *number of units purchased from Supplier Y* that are jointly consistent with the given information. Make only two selections, one in each column.

Number of units purchased from Supplier X	Number of units purchased from Supplier Y	
		15
		20
		22
		25
		27
		30

Answer Explanation:

Strategize

If x represents the number of units purchased from Supplier X and y represents the number of units purchased from Supplier Y, then from the given information,

$$\begin{aligned} \frac{(100 + 20(10) + 5(x - 10) + 150 + 15(10) + 10(y - 10))}{(x + y)} &= 20 \\ 450 + 5x + 10y &= 20x + 20y \\ 450 &= 15x + 10y \\ 90 &= 3x + 2y \end{aligned}$$

The value of x must be even because if x is odd, then $3x + 2y$ cannot equal an even number. Therefore, $x = 20$ or $x = 22$ or $x = 30$. If $x = 20$, then $2y = 30$ and $y = 15$.

If $x = 22$, then $2y = 24$ and $y = 12$. If $x = 30$, then $2y = 0$ and $y = 0$. Of these three possible values for y, only 15 is among the answer choices.

Number of units purchased from Supplier X:

As shown above, the company purchased 20 units of the product from Supplier X.

The correct answer is *20*. *Number of units purchased from Supplier Y*:

As shown above, the company purchased 15 units of the product from Supplier Y.

The correct answer is *15*.

•

A financial adviser was showing a client the value, rounded to the nearest cent, of an initial investment of \$100.00 after 5, 10, 15, 20, 25, and 30 years, under the assumption that the value increases by r% per year for some positive constant r. The adviser correctly gave \$130.01 for the value after 5 years. However, the adviser inadvertently made two transcription errors, and as a result, two of the remaining values shown to the client were incorrect.

Assuming the options provided are the remaining five amounts shown to the client, select for *First error* the lesser of the two incorrectly transcribed values, and select for *Second error* the greater of the two incorrectly transcribed values. Make only two selections, one in each column.

First error	Second error	
		\$160.02

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		$219.75
		$285.70
		$317.43
		$482.90

Answer Explanation:

Strategize

It is given that after 5 years, the value of the investment was $130.01. It follows that $100\left(1+\frac{r}{100}\right)^5 = 130.01$, so $\left(1+\frac{r}{100}\right)^5 = 1.3001$. After 10 years, the value of the investment is 100

$$\left(1+\frac{r}{100}\right)^{10} = 100\left(\left(1+\frac{r}{100}\right)^5\right)^2 = 100(1.3001)^2 = 130.01(1.3001) = 169.02.$$

After 15 years, the value of the investment is 100

$$\left(1+\frac{r}{100}\right)^{15} = 100\left(\left(1+\frac{r}{100}\right)^5\right)^3 = 100(1.3001)^3 = 169.02(1.3001) = 219.75.$$

After 20 years, the value of the investment is 100

$$\left(1+\frac{r}{100}\right)^{20} = 100\left(\left(1+\frac{r}{100}\right)^5\right)^4 = 100(1.3001)^4 = 219.75(1.3001) = 285.70.$$

After 25 years, the value of the investment is 100

$$\left(1+\frac{r}{100}\right)^{25} = 100\left(\left(1+\frac{r}{100}\right)^5\right)^5 = 100(1.3001)^5 = 285.70(1.3001) = 371.44.$$

After 30 years, the value of the investment is 100

$$\left(1+\frac{r}{100}\right)^{30} = 100\left(\left(1+\frac{r}{100}\right)^5\right)^6 = 100(1.3001)^6 = 371.44(1.3001) = 482.90.$$

***First error*:**

The value of the investment after 10 years was $169.02, but the financial adviser gave $160.02. This was the financial adviser's first error.

The correct answer is $*160.02*.

***Second error*:**

The value of the investment after 25 years was $371.43, but the financial adviser gave $317.43. This was the financial adviser's second error.

The correct answer is $*317.43*.

•

A clothing retailer used to sell only "fast-fashion" pieces, which were low priced and had a profit markup of 50 percent of the per-item cost (including, e.g., the costs of wholesale purchase and marketing). On average, each customer spent $850 annually on around 65 such pieces from the retailer. Now the retailer wishes to double its total profits by selling only "classic" pieces. It plans to double its percentage profit markup per item and generate more revenue per customer while leaving unchanged the company's total costs. The plan assumes that for each classic piece, on average, customers will pay five times what they paid for each fast-fashion piece, and that the total number of customers for the retailer's clothing products will remain the same.

Statements: Customers paid an average of ____1____ dollars (rounded to the nearest dollar) for each of the retailer's fast-fashion pieces. The retailer will need to sell an average minimum of ____2____ classic pieces per person

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fast-fashion pieces. The retailer will need to sell an average minimum of ____2____ classic pieces per person (rounded to the nearest whole number) to achieve its profit goals for classic pieces.

Select values for *1* and for *2* that create the statements that are most strongly supported by the information provided and in accordance with the retailer's plan. Make only two selections, one in each column.

1	2	
		10
		13
		15
		17
		21
		25

Answer Explanation:

Strategize

***1*:**

Customers paid an average of $\frac{\$850}{65} \approx \$13.07$ or \$13 rounded to the nearest dollar.

The correct answer is *13*.

***2*:**

For fast-fashion items, let \$C represent the cost per item. For these items, there is a 50% markup on the per-item cost. The customer pays an average of $\frac{\$850}{65}$ per item, so $\frac{\$850}{65} = 1.5(\$C) = \frac{3}{2}(\$C)$. It follows that the retailer's per-item cost \$C is $\frac{2}{3}\left(\frac{\$850}{65}\right)$. The retailer's profit per item is $\frac{\$850}{65} - \frac{2}{3}\left(\frac{\$850}{65}\right) = \frac{1}{3}\left(\frac{\$850}{65}\right)$. The retailer's total profit per customer is $65\left(\frac{1}{3}\left(\frac{\$850}{65}\right)\right) = \frac{\$850}{3}$. For classic items, let \$c be the per-item cost. The retailer plans to double the percentage profit markup of the cost to generate revenue of 2(\$c) per item. However, the customer is willing to pay $5\left(\frac{\$850}{65}\right)$, so the retailer's cost per item, \$c, is $\frac{5}{2}\left(\frac{\$850}{65}\right)$. It follows that the retailer's per-item profit is $5\left(\frac{\$850}{65}\right) - \frac{5}{2}\left(\frac{\$850}{65}\right) = \frac{5}{2}\left(\frac{\$850}{65}\right)$.

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